DOB-MPC for Robust Trajectory Tracking of Autonomous Surface Vehicles

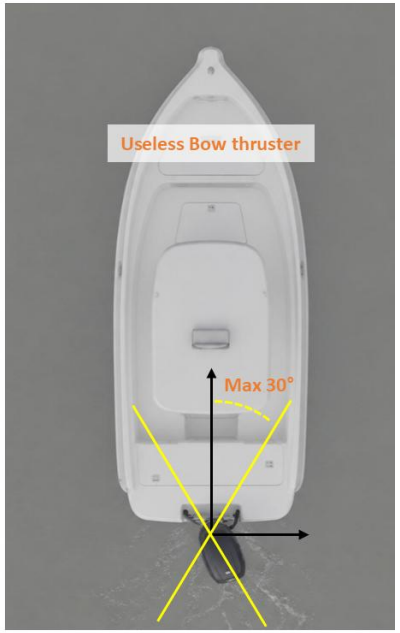
Mobile Robotics & Intelligence Lab.

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Introduction

Challenges of ASV control



< Underactuation >



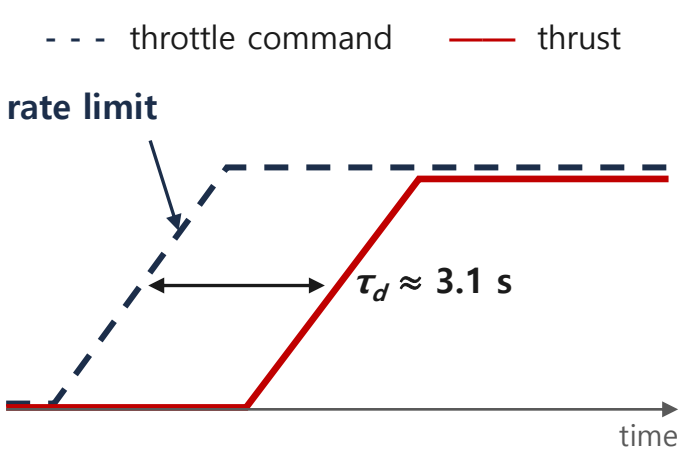
< Environmental disturbance >



diesel engine



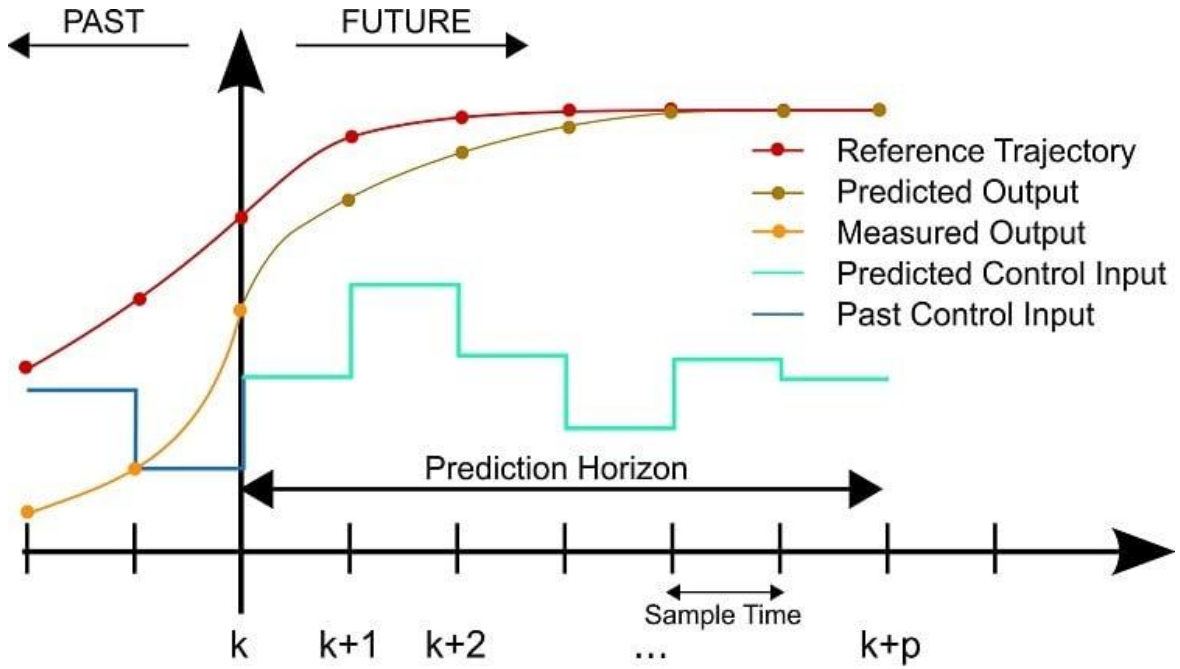
servo actuator



< Engine dynamics >

Background

Model Predictive Control (MPC)



< Receding-horizon prediction >

$$\min_{\mathbf{x}(\cdot),\mathbf{u}(\cdot)} \quad \sum_{i=0}^{N-1} \ell\left(\mathbf{x}_i, \mathbf{r}_i, \mathbf{u}_i\right) + \ell_T\left(\mathbf{x}_N, \mathbf{r}_N\right),$$

$$\text{s.t.} \quad \mathbf{x}_0 - \mathbf{x}_{\text{init}} = 0,$$

$$\mathbf{x}_{i+1} - f_d\left(\mathbf{x}_i, \mathbf{u}_i\right) = 0, \quad i = 0, \ldots, N-1,$$

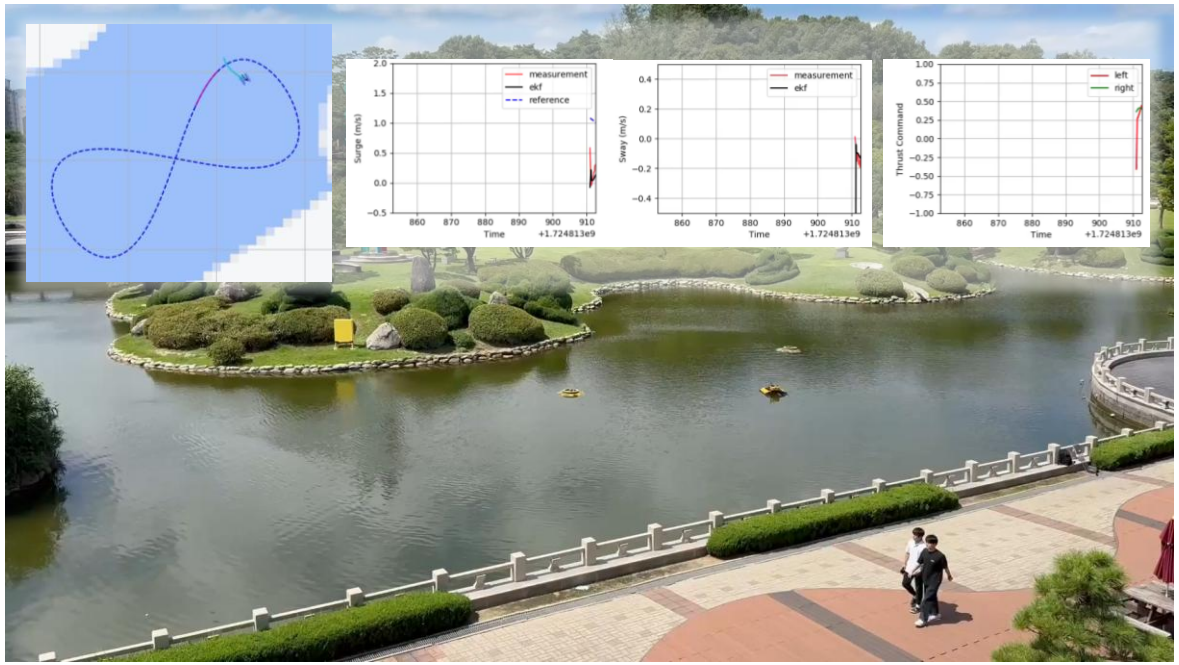
$$g\left(\mathbf{u}_i\right) \leq 0, \quad i = 0, \ldots, N-1,$$

< MPC optimal control problem >

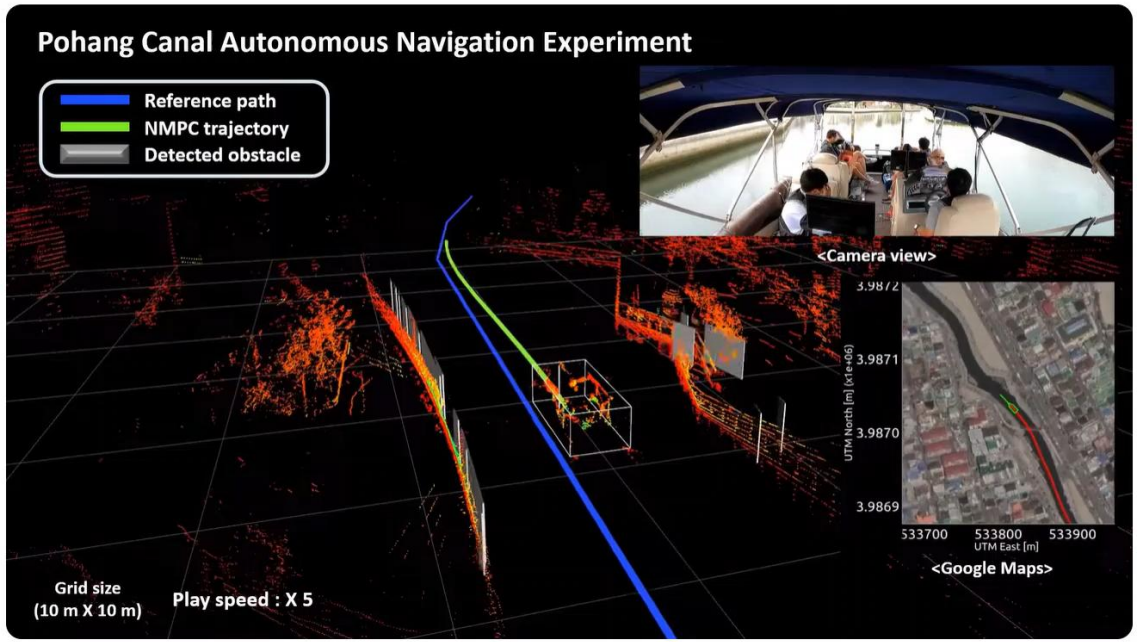
- Predict future states over a horizon
- Optimize inputs subject to constraints
- Naturally handles input constraints & underactuation

Background

Prior work — ASV control with MPC



< Duck-pond trajectory tracking >



< Canal navigation >

MPC works well in calm environment → tracking degrades under disturbance & model uncertainty

DOB-MPC

Control structure

Objective

- Make the error between system states and reference trajectory 0

$$J(t) = \sum_{i=0}^{N-1} (\| \eta_{t+i} - \mathrm{r}_{t+i} \|_Q^2 + \| \Delta \mathrm{u}_{t+i} \|_P^2) + \| \eta_{t+N} - \mathrm{r}_{t+N} \|_R^2$$

"state" *"global path"* *"Input efforts"*

$$\eta = [x, y, \psi, u, v, r]^T$$

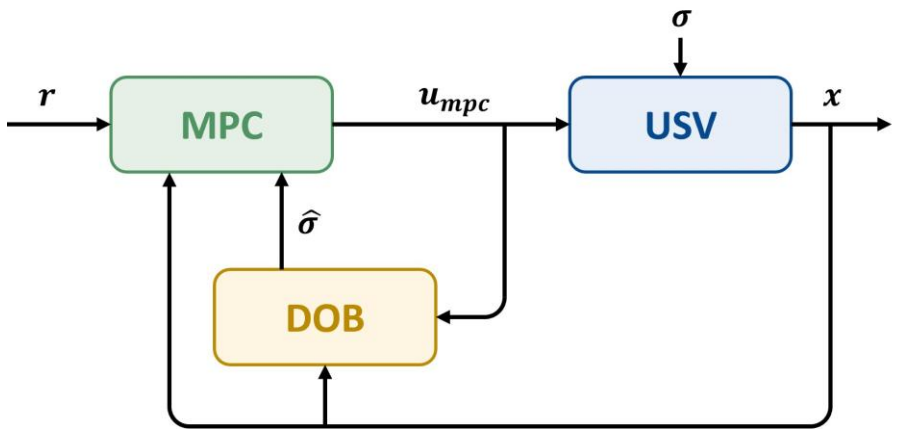
Subject to

$$\mathrm{x}_{k+1} = f(\mathrm{x}_k, \mathrm{u}_k) + \mathrm{w}_k$$

"Estimated disturbance"

$$\mathrm{F}_{min} \le \mathrm{F} \le \mathrm{F}_{max} \qquad \dot{\mathrm{F}}_{min} \le \dot{\mathrm{F}} \le \mathrm{F}_{max}$$

$$\delta_{min} \le \delta \le \delta_{max} \qquad \dot{\delta}_{min} \le \dot{\delta} \le \dot{\delta}_{max}$$



< Control diagram >

DOB-MPC

Control structure

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- Make the error between system states and reference trajectory 0

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$$\delta_{min} \le \delta \le \delta_{max} \qquad \dot{\delta}_{min} \le \dot{\delta} \le \dot{\delta}_{max}$$

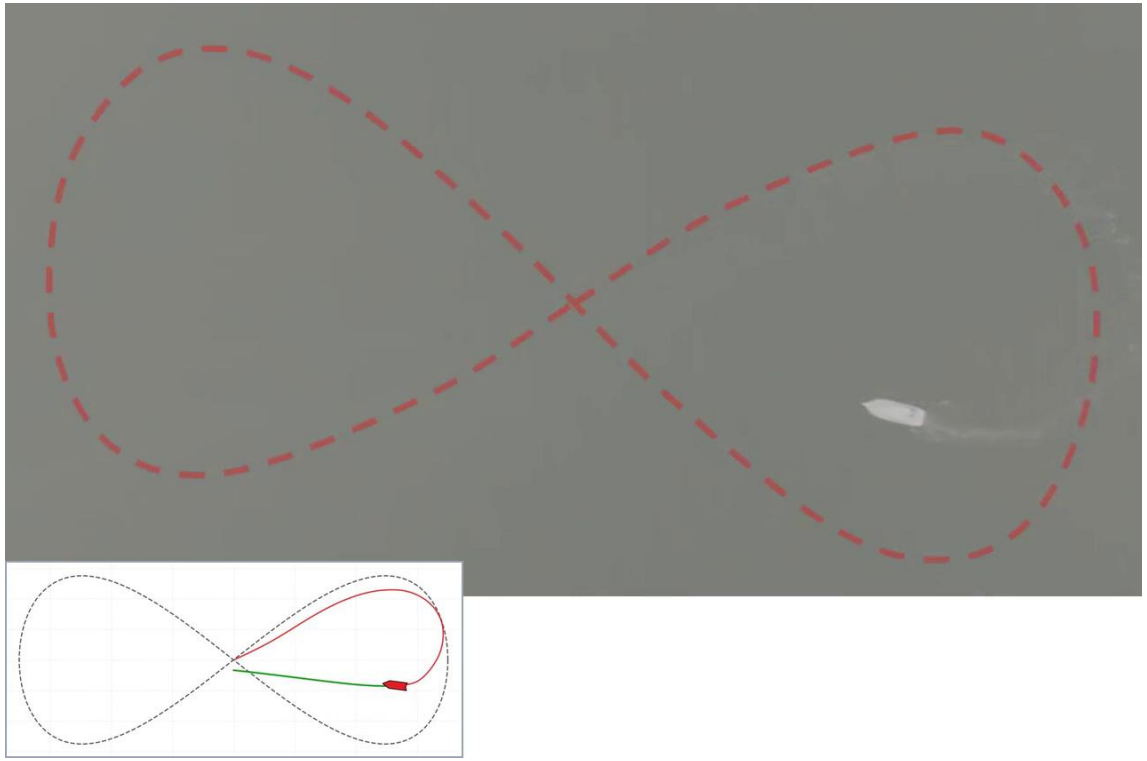


< AURA ASV >

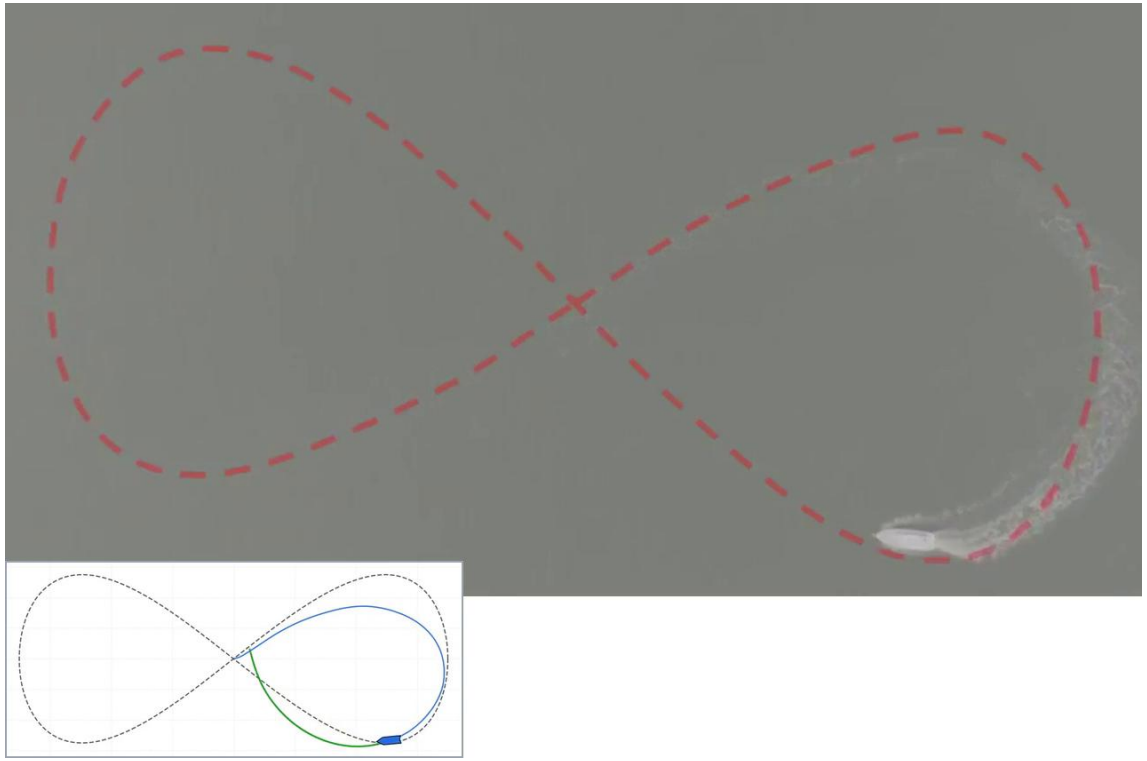
DOB-MPC

Representative runs (Drone view) (110 m × 55 m)

Nominal MPC



DOB-MPC (proposed)



Thank you!